Accessibility improved: 96 to 100

96 / 100	100 / 100	+4 points
BEFORE	AFTER	

Project: Memory Match - Neon Arcade. Category: Lighthouse Accessibility.
Page: https://sudip09-sys.github.io/WebProgramming/teamproject/index.html
Initial report: Mobile emulation (Moto G Power), Lighthouse 13.5.0, Chromium 156.

Three improvements implemented

1. Correct board semantics - automated failure fixed

Added role="group" to the card-board div so its aria-label is permitted.
The expanded initial audit identifies div#board.board as the failing element.

2. Add keyboard skip navigation - manual improvement

Added a first-in-body skip link, a focusable main target, and a visible focus style.
Keyboard users can bypass repeated header links and move directly to main content.

3. Focus the game result - manual improvement

When a round ends, focus moves once to the result heading so the result and replay control can be reached. The one-time flag resets when another round starts.

Evidence integrity: the initial report had ONE failing automated check, not three.
The other two changes address keyboard accessibility. A score of 100 does not establish full WCAG conformance. Interactive keyboard and screen-reader checks remain pending.

Before and after code

1 / Card board role - teamproject/index.html

```
BEFORE: <div id="board" class="board" aria-label="Memory cards"></div>
AFTER:  <div id="board" class="board" role="group"
             aria-label="Memory cards"></div>
```

2 / Skip navigation - index.html and style.css

```
BEFORE: Body started with header; main had no skip target.
AFTER:  <a class="skip-link" href="#main">Skip to main content</a>
        <main id="main" tabindex="-1">
        .skip-link { position:absolute; z-index:1000;
                     transform:translateY(-150%); }
        .skip-link:focus { transform:translateY(0);
                           outline:3px solid #90e7d4; }
```

3 / Focus new results - index.html and game.mjs

```
BEFORE: Result became visible without an explicit focus transfer.
AFTER:  <h2 id="result-title" tabindex="-1"></h2>
        // Initialize and reset for each round: announcedResult = false;
        if (!announcedResult) {
          announcedResult = true; $('result-title').focus();
        }
```

Validation and remaining checks

JavaScript syntax check passed (node --check); GitHub Pages deployment succeeded.

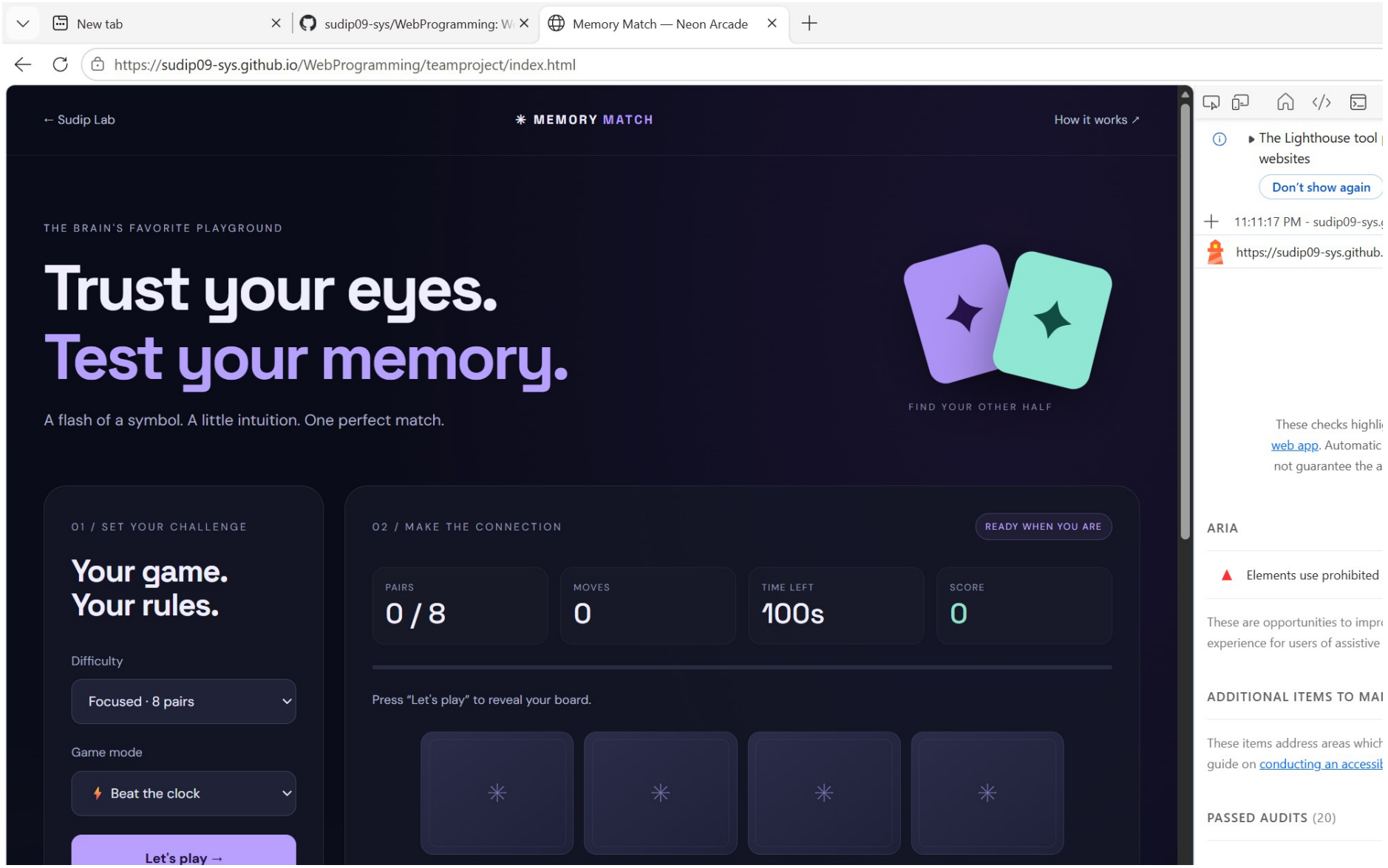
Follow-up score: 100/100, captured by the student. Manual keyboard testing remains pending:

Tab to the skip link and activate it; finish a round and check heading/replay focus.

Commit: e7e5d0004fdb2a73f4a91c693f07f1f8334bf9af

Before / score: 96 out of 100

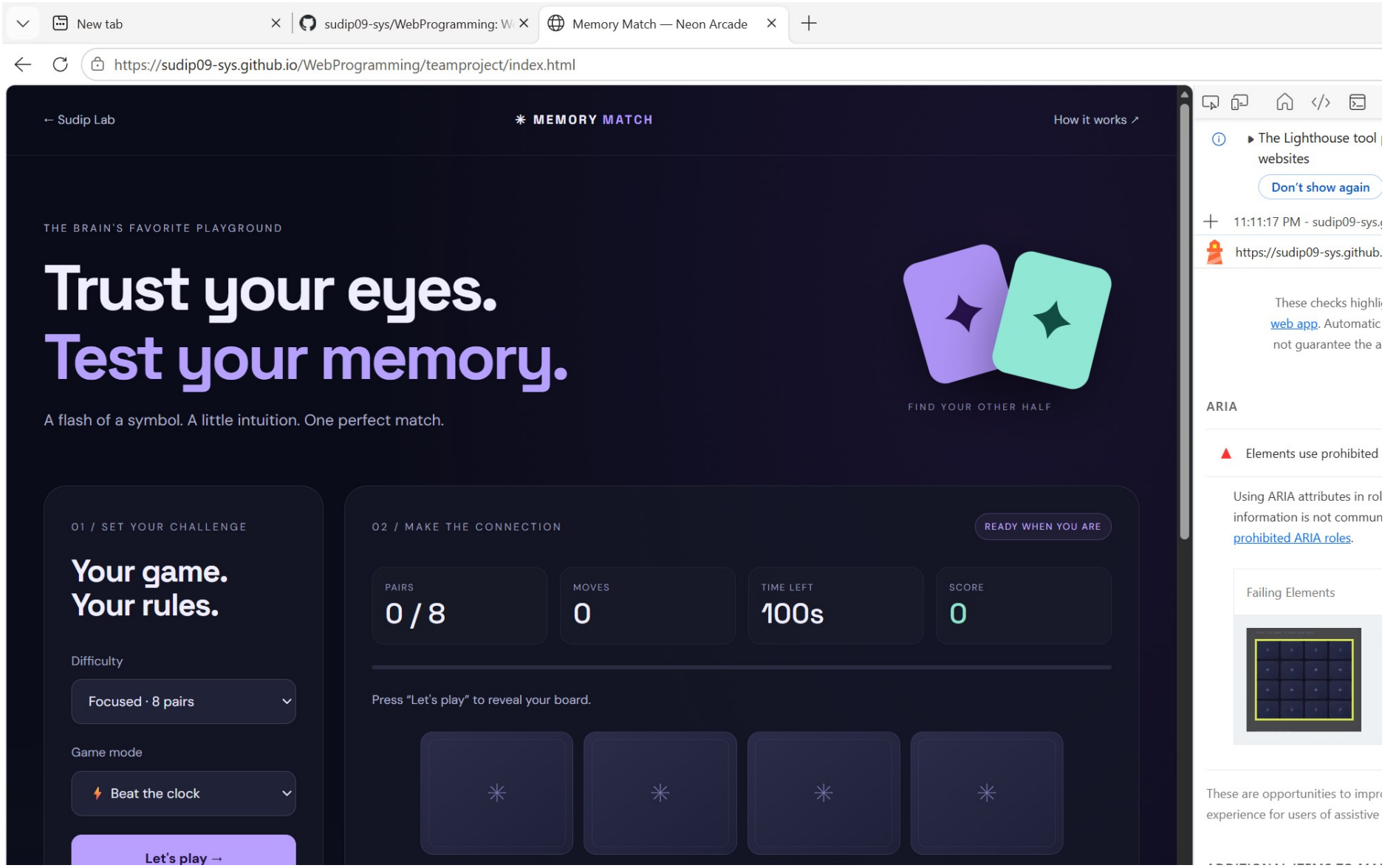
Initial report: one automated failure - Elements use prohibited ARIA attributes.



Student-supplied screenshot: Screenshot 2026-10-28 231322.png. Original report evidence.

Before / failing element identified

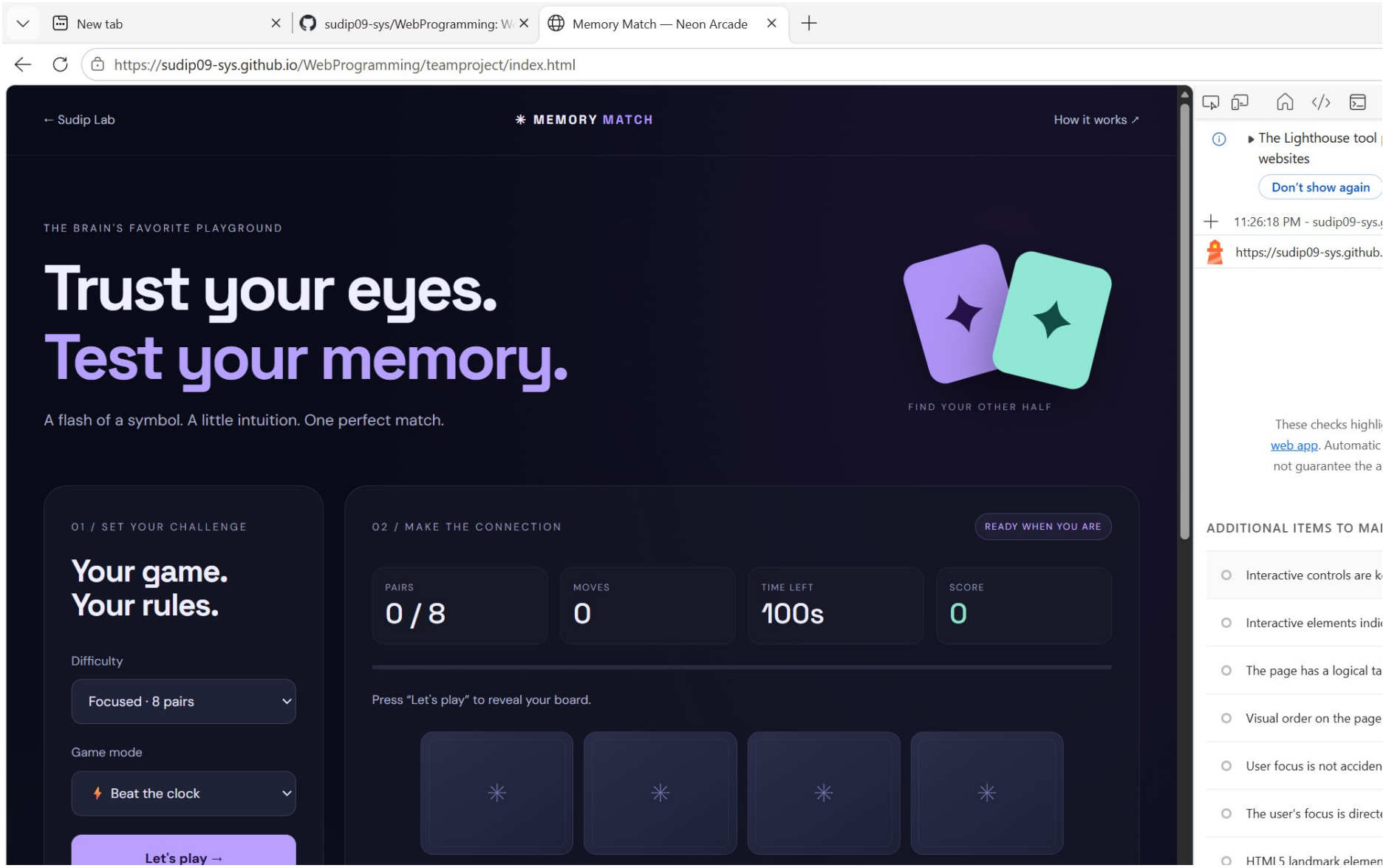
Expanded audit identifies div#board.board as the affected element.



Student-supplied screenshot: Screenshot 2026-10-28 232321.png. Original report evidence.

After / score: 100 out of 100

Follow-up report captured after the three improvements were deployed.



Student-supplied screenshot: Screenshot 2026-10-28 232658.png. Original report evidence.